ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

INTRODUCTION TO PROGRAMMING

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 1st year, semester 1 final examination (Python 3): data types and operators, control structures, functions, lists and strings, tracing and debugging, files, and an introduction to object orientation (classes).

Written by Academic Central for practice, following first-year Introduction to Programming syllabuses for engineering students at South African universities (Python). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. All code must be written in Python 3. Use meaningful variable names and indent your code clearly.
3. Small syntax mistakes that do not change the logic will not be penalised in QUESTIONS 5 and 6.
4. No computers or calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer and write only the letter (A–D) next to the question number, for example 1.11 B.

1.1 Which one of the following is a valid Python variable name? (1)

A 2speed
B motor_speed
C motor-speed
D class

1.2 What is the value of 7 // 2? (1)

A 3.5
B 3
C 4
D 1

1.3 What is the value of 7 % 3? (1)

A 2
B 0
C 1
D 3

1.4 What does type(3.0) return? (1)

A int
B float
C str
D bool

1.5 What is len([1, [2, 3], 4])? (1)

A 4
B 2
C 3
D 1

1.6 Which keyword is used to define a function? (1)

A func
B def
C function
D void

1.7 What is the value of bool("")? (1)

A True
B False
C None
D It causes an error

1.8 Which values does range(2, 10, 3) produce? (1)

A 2, 5, 8
B 2, 5, 8, 11
C 3, 6, 9
D 2, 4, 6, 8

1.9 Which data type cannot be changed after it is created? (1)

A list
B dict
C tuple
D set

1.10 What happens when an existing file is opened with mode "w"? (1)

A New text is added to the end
B The file can only be read
C The file is emptied and writing starts at the beginning
D An error occurs

[10]

QUESTION 2: PROGRAMMING CONCEPTS

2.1 Explain the difference between a compiler and an interpreter. (2)

2.2 Explain the difference between a syntax error, a runtime error and a logic error. Give a short Python example of each. (6)

2.3 What is an algorithm? Give TWO properties that a good algorithm must have. (3)

2.4 Give TWO reasons why programmers divide a program into functions. (2)

2.5 Explain the difference between a local variable and a global variable. (2)

[15]

QUESTION 3: TRACING CODE

Study each piece of code and answer the question. Show your working where it helps you.

3.1 What is the output of the code below? (2)

```
x = 5
y = 2
x = x + y * 3
y = x % 4
print(x, y)
```

3.2 What is the output of the code below? (4)

```
total = 0
for i in range(1, 6):
    if i % 2 == 0:
        total += i
    else:
        total -= 1
print(total)
```

3.3 What is the output of the code below? (3)

```
n = 20
count = 0
while n > 1:
    n = n // 2
    count += 1
print(n, count)
```

3.4 What is the output of the code below? (3)

```
def scale(v, k=2):
    return v * k

readings = [3, 8, 1]
out = [scale(r) for r in readings if r > 2]
print(out, scale(4, 3))
```

3.5 Draw a trace table with the columns i, values[i], big and pos for the code below, and give the output. (5)

```
values = [4, 7, 2, 9, 5]
big = values[0]
pos = 0
for i in range(1, len(values)):
    if values[i] > big:
        big = values[i]
        pos = i
print(big, pos)
```

3.6 What is the output of the code below? (3)

```
s = "ROBOT"
print(s[1:4], s[::-1], s.lower().count("o"))
```

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QUESTION 4: DEBUGGING

The function below should return the average of a list of temperature readings, and the last line should print it. It contains FOUR problems.

```
def average(temps):
    total = 0
    for t in temps
        total = total + t
    return total / len(temp)

print("Average: " + average([20, 22, 24]))
```

4.1 Identify the FOUR problems. For each one, say whether it is a syntax error, a runtime error or a possible runtime error, and explain how to fix it. (Hint: one problem only happens when the list is empty.) (8)

4.2 Rewrite the code so that it works correctly. (2)

[10]

QUESTION 5: WRITING CODE

5.1 Write a function c_to_f(c) that returns the temperature c (in °C) converted to °F, using F = C × 9/5 + 32. (3)

5.2 A distance sensor gives readings in cm. Write a program that asks the user to type readings one at a time until they type −1. The program must then print: the number of readings, the smallest reading, and how many readings were below 10 cm ("too close"). Assume at least one reading is typed. (8)

5.3 Write a function is_prime(n) that returns True if the whole number n is a prime number and False if it is not. (5)

5.4 Write a function motor_state(speed) that returns "ERROR" if speed is negative, "STOP" if speed is 0, "SLOW" if speed is more than 0 and at most 30, and "FAST" if speed is more than 30. (4)

5.5 Write a function moving_average(data) that returns a new list with the average of every 3 neighbouring values. For example, moving_average([1, 2, 3, 4]) returns [2.0, 3.0]. (5)

5.6 Write a class Motor with: an __init__ method that stores the motor's name and sets its speed to 0; a method set_speed(s) that changes the speed only if s is from 0 to 100 (otherwise it prints "Invalid speed"); and a method stop() that sets the speed to 0. Then create a Motor called "left wheel" and set its speed to 60. (5)

[30]

QUESTION 6: WORKING WITH FILES

A data logger saves a file called log.txt. Each line has a time and a temperature, separated by a comma:

```
08:00,21.5
09:00,23.0
10:00,26.5
11:00,24.0
```

6.1 Write code that reads log.txt and stores the times in one list and the temperatures (as floats) in another list. (5)

6.2 Write code that prints the time at which the highest temperature was recorded. (3)

6.3 Write code that writes every reading above 24.0 to a new file alarm.txt, one line per reading, in the same format as log.txt. (4)

6.4 Why is it good practice to open files with a with statement? (1)

6.5 What happens if log.txt does not exist? Show how your program can handle this without crashing. (2)

[15]

TOTAL: 100

END OF QUESTION PAPER